

WALNUT RIDGE, AR, USA

WMO: 723406

Lat: 36.125N

Lon: 90.924W

Elev: 273

StdP: 14.55

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93991

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.7	19.1	1.4	6.0	17.0	7.1	8.0	22.3	27.3	48.8	24.9	47.6	9.2	0	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	95.2	77.5	92.9	76.8	90.9	76.2	80.5	91.0	79.2	89.7	78.1	88.0	9.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	144.7	86.3	75.8	136.5	84.9	75.0	132.5	84.0	44.3	91.2	42.8	90.0	41.6	88.9	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.8	18.7	8.8	99.7	3.5	3.1	6.3	101.9	4.2	103.8	2.2	105.5	-0.3	107.8
			7.5	82.5	3.5	1.5	4.9	83.6	2.9	84.5	0.9	85.3	-1.7	86.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.0	37.6	41.5	50.4	60.1	69.6	78.0	80.4	78.9	71.8	61.4	49.3	40.3	
	DBStd	17.19	10.49	10.23	9.62	8.08	7.18	5.09	4.56	4.68	6.65	8.42	8.93	9.66	
	HDD50	1261	404	271	114	15	1	0	0	0	0	10	120	326	
	HDD65	3626	849	658	460	185	41	1	0	0	19	173	475	765	
	CDD50	4925	20	34	126	319	609	840	942	895	654	362	98	26	
	CDD65	1818	0	0	7	40	184	391	477	431	223	60	4	1	
	CDH74	17497	0	3	45	308	1475	3981	5017	4169	1922	545	31	1	
	CDH80	7244	0	0	2	53	460	1745	2265	1840	735	141	3	0	
Wind	WSAvg	8.4	9.7	9.9	10.4	10.6	8.7	7.5	6.5	5.9	6.3	7.4	8.7	8.9	
Precipitation	PrecAvg	47.2	3.5	3.4	4.3	4.8	4.7	3.1	3.8	3.1	3.5	3.7	4.5	4.3	
	PrecMax	63.3	7.2	8.2	11.1	11.1	7.9	6.5	7.4	7.8	8.3	11.5	10.8	9.0	
	PrecMin	35.2	1.0	1.1	1.5	0.8	0.8	1.4	1.0	0.2	0.1	0.0	1.0	0.3	
	PrecStd	7.2	1.6	1.7	2.0	2.2	2.1	1.3	1.9	1.7	1.9	2.3	2.8	2.2	
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.3	72.3	79.1	84.3	91.2	97.1	99.1	98.7	93.6	88.3	79.1	68.5
MCWB			62.8	63.5	65.9	68.0	72.6	76.8	78.1	79.2	74.8	72.3	65.3	62.3	
2%		DB	63.3	66.1	74.3	81.5	88.3	93.3	95.3	94.6	90.8	83.8	72.2	63.6	
		MCWB	59.1	59.6	62.7	66.6	72.2	75.5	78.3	78.3	74.6	69.5	61.5	59.3	
5%		DB	58.8	62.9	70.2	78.6	84.4	91.2	92.8	91.3	88.2	80.8	67.6	59.5	
		MCWB	53.7	55.9	60.5	64.7	70.9	74.9	78.0	77.2	73.6	66.2	60.6	54.8	
10%		DB	54.2	57.3	65.8	73.8	81.9	89.7	90.6	89.8	83.9	75.5	63.6	54.9	
		MCWB	49.5	51.6	58.0	62.4	69.3	74.6	77.5	76.7	71.3	63.9	56.6	50.8	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures		0.4%	WB	63.6	65.1	67.9	71.8	76.7	80.7	82.3	82.3	78.5	75.3	68.2	63.9
			MCDB	65.6	70.3	76.2	80.8	85.9	91.1	92.2	92.6	89.4	84.6	74.1	67.1
	2%	WB	60.3	60.8	65.2	69.1	74.9	78.5	80.6	80.7	77.0	71.7	64.3	60.4	
		MCDB	62.9	65.3	72.5	78.0	83.5	89.3	91.0	91.1	87.4	80.1	69.1	63.5	
	5%	WB	55.0	57.4	62.2	66.8	73.1	77.2	79.3	79.1	75.4	69.2	61.4	55.8	
		MCDB	57.5	61.6	68.0	75.0	82.0	87.8	89.8	89.6	84.6	76.3	66.5	58.8	
	10%	WB	49.6	52.5	58.9	64.6	71.2	75.9	78.3	77.8	73.7	66.6	57.9	51.6	
		MCDB	54.1	57.6	64.6	72.0	79.9	86.0	88.1	87.1	81.1	73.6	63.2	54.9	
	Mean Daily Temperature Range	5% DB	MDBR	15.6	16.9	18.9	19.9	18.6	18.5	17.5	18.7	20.8	21.9	18.2	15.1
			MCDBR	20.7	23.9	23.6	23.4	21.6	20.7	19.8	21.3	23.6	25.8	24.1	19.7
MCWBR			17.0	17.9	15.0	12.7	10.1	7.9	7.4	9.4	10.5	13.1	15.9	16.0	
5% WB		MCDBR	17.8	20.6	20.4	19.2	18.4	18.3	18.3	19.4	20.1	21.5	20.2	17.5	
		MCWBR	16.8	17.8	14.6	12.4	9.8	8.3	7.9	9.3	9.8	12.1	15.6	16.1	
Clear-Sky Solar Irradiance	taub		0.316	0.322	0.357	0.401	0.437	0.459	0.486	0.476	0.413	0.355	0.324	0.310	
	taud		2.476	2.458	2.392	2.268	2.229	2.181	2.146	2.153	2.322	2.469	2.505	2.510	
	Ebn at Noon		279	291	289	280	270	263	255	255	267	276	274	275	
	Edh at Noon		27	30	36	42	45	47	48	47	38	30	26	24	
All-Sky Solar Radiation	RadAvg		756	936	1289	1639	1820	2078	1995	1821	1544	1173	824	644	
	RadStd		67	121	102	113	143	133	96	111	107	137	88	71	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page